

Roll No.

7575

B. C. A. (Second Semester)

EXAMINATION, 2025-26

Paper III

DISCRETE MATHEMATICS

Course Code : B230203T

Time : Two Hours]

[Maximum Marks : 75

Note : This question paper is divided into three Sections—A, B and C. Attempt questions from all Sections as directed.

Inst. : The candidates are required to answer only in serial order. If there are many parts of a question, answer them in continuation.

Section—A

Very Short Answer Type Questions

Note : Attempt any *three* questions. Give answer of each question in about **50** words. Each question carries 3 marks. $3 \times 3 = 9$

1. (a) What is a proposition ? Give examples.
- (b) What are quantifiers in predicate logic ?
- (c) What do you mean by reflexive relation ? Explain.
- (d) What is a contradiction ? Prove that $(p \rightarrow q) \wedge (p \wedge \neg q)$ is a contradiction.
- (e) Draw a complete bipartite graph for $K_{2,3}$ and $K_{2,2}$.

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Section—B

Short Answer Type Questions

Note : Attempt any *four* questions. Answer each question in about 225 words. Each question carries 9 marks. $4 \times 9 = 36$

2. (a) Test the validity of the following argument : "If two sides of a triangle are equal then the opposite angles are equal. Two sides of a triangle are not equal therefore the opposite angles are not equal."
- (b) State and prove De Morgan's Law.
- (c) What is a directed graph ? Explain Adjacency Matrix and Incidence Matrix with the help of a directed graph.
- (d) What do you mean by an equivalence relation ? Find whether the relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$ on the set $A = \{1, 2, 3\}$ is an equivalence relation or not.
- (e) Write the Distributive law for logical equivalences. Prove that $(p \rightarrow q) \wedge (q \rightarrow r) \equiv p \rightarrow (q \wedge r)$.
- (f) Explain functionally complete set of connectives with example.
- (g) Write a short note on connectedness of a directed graph.

Section—C

Long Answer Type Questions

Note : Attempt any *two* questions. Answer each question in about 475 words. Each question carries 15 marks. $2 \times 15 = 30$

3. (a) Among 100 students, 32 study Mathematics, 20 study Physics, 45 study Biology, 15 study Mathematics and Biology, 7 study Mathematics and Physics, 10 study Physics and Biology and 5 study all the three subjects.

Find :

- (i) The number of students who study at least one of the three subjects.

(ii) The number of students who do not study any of the three subjects.

(iii) The number of students who study only Physics.

(b) Explain Warshall's algorithm for transitive closure. Find the transitive closure of R using Warshall's algorithm where $R = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$ on the set $A = \{1, 2, 3\}$.

(c) What are normal forms ? Convert the following expression into DNF and CNF : $(p \leftrightarrow q)$.

(d) Write short notes on the following terms :

Degree of vertex in a graph, Domain and range, Tautology, Asymmetric relation.

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